Business Process Engineering
Week # 9

SPRING-2024

Lean 101

An introduction to Lean principles, methodology, tools and terminology

What is Lean?

- A time-tested method and set of tools to help us improve “how” we produce our products and services.
- Lean is also a mindset, where we ask each day “How can we make our services better for customers?”

The Lean Process

- A method for creating a more effective business by eliminating wasteful practices and improving efficiency
- Core idea is to maximize customer value while minimizing waste
- By reducing waste and improving customer satisfaction, the lean process saves companies money and improves overall profitability
- Lean is a way of thinking and acting for an entire organization
- Businesses in all industries and services, including healthcare and governments, are using lean principles as the way they think and do

Lean helps us Understand:

1. What adds value to our customers
2. How work gets done
3. How we can identify root causes of problems
4. What an “ideal” process looks like
5. How we can improve performance
6. Whether process changes were successful

Lean is about Simplifying our Work

- Eliminate tasks that do not add value
- Make things easy and intuitive for customers and staff
- Automate repetitive tasks
- Leverage staff talent

How Do We Define Value-added?

Value-added **vs.** **Non Value-added**

- Customer is willing to pay for it
- Actually transforms a product or service

- Consumes resources without creating value for the customer
- Requires extra time, effort, or resources

Lean is NOT…

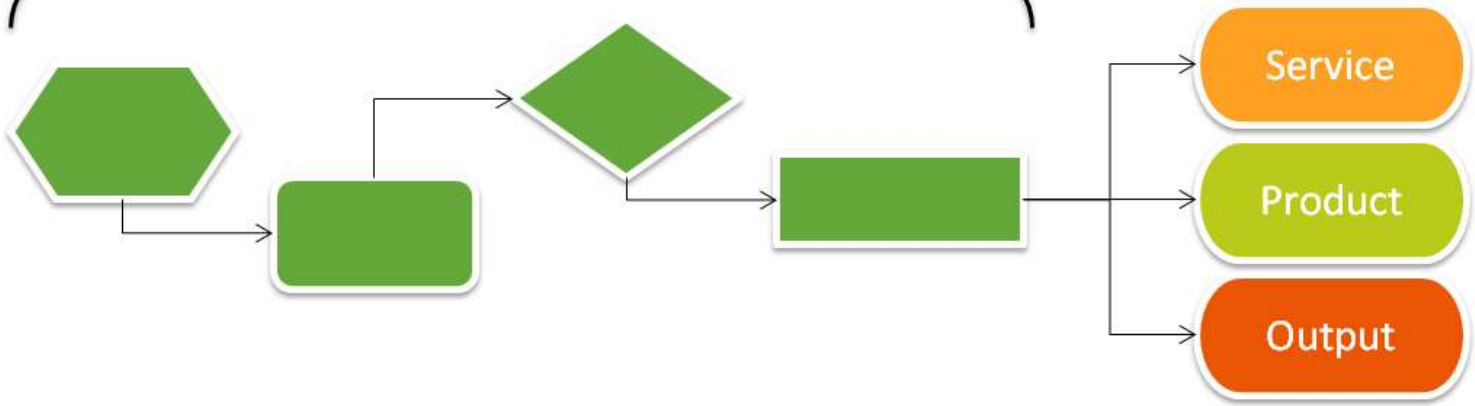
- Not an acronym (LEAN)
- Not a solution to personnel or performance issues
- Not an initiative to reduce headcount – it's about improving service
- Not a replacement for Six Sigma – it is complementary
- Not a "manufacturing thing"

Lean does NOT require special expertise

Why Focus on Process?

85%

of improvement opportunities are here

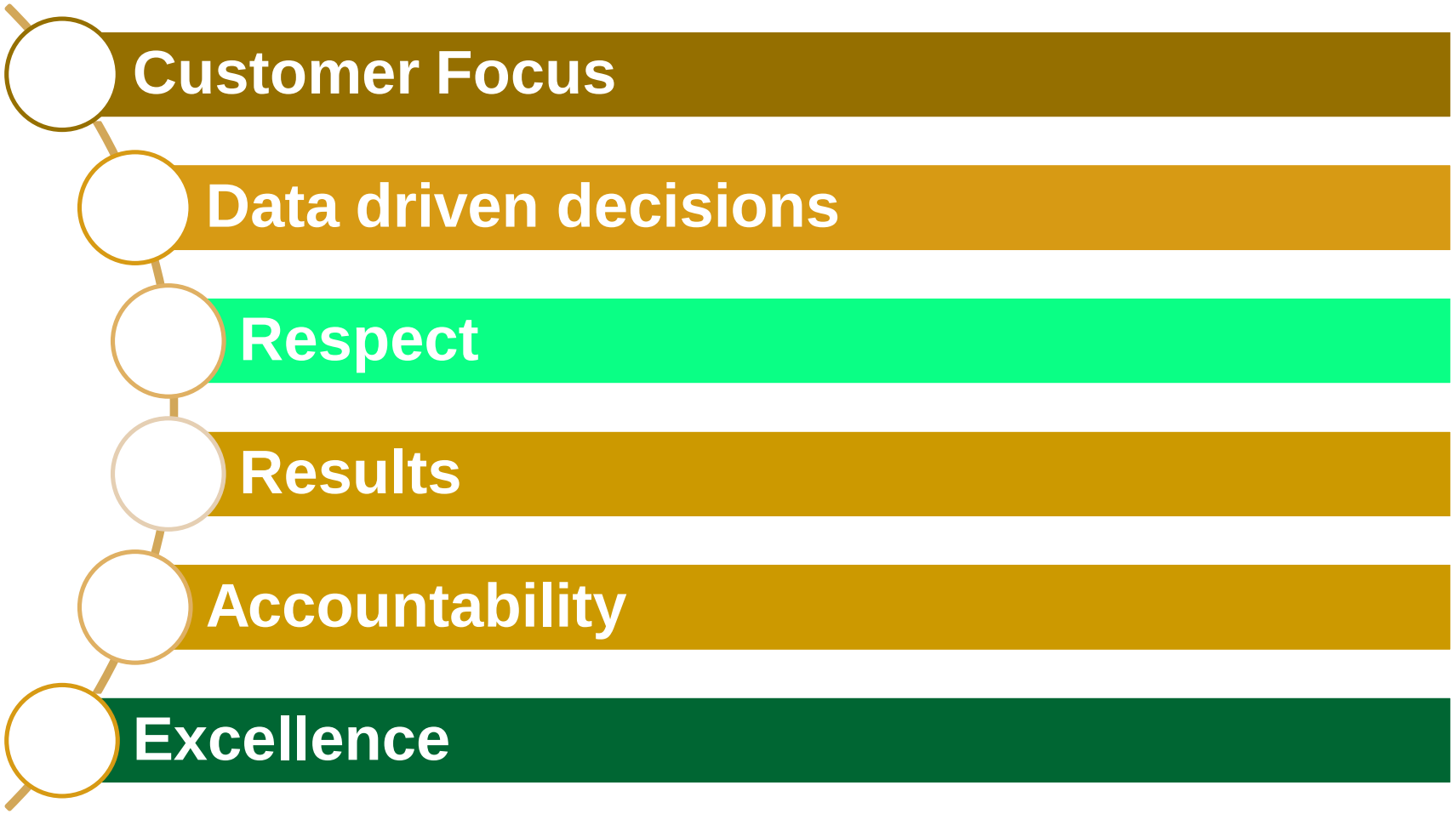


Why Lean?

- Increasing customer expectations
- Pressure for greater accountability and transparency
- Tight and shrinking budgets
- Shrinking workforce and increasing need for a more skilled workforce.

Lean helps us improve quality, reduce costs, increase customer and employee satisfaction, & capture knowledge

Lean Principles



Principle 1: Customer Focus

Better, faster, cheaper…

Accurate	Timely	Accessible	Treatment	Effective
- What I want	- When I want	- How I want (Easy to use)	- Feel my needs are understood and that I am treated fairly and with respect	- Service achieves desired results

Cost
Right price or resource investment

Principle 2: Data Driven Decisions

- Verify anecdotes and feelings with data!
- Complaints that a process doesn't work or is too slow?
- Gather data to confirm!
- Difficulty deciding which solution will work best?
- Test, make decision based on data!

Principle 3: Respect



"A bad process
will beat
a good person
every time"

- W. Edwards Deming

It's about the Process

Principle 4: Results

Set SMART goals and measure results

- Specific
- Measurable
- Attainable (challenging, but within reach)
- Relevant (aligned with your strategic priorities)
- Time-bound
  - Example: Reduce the time it takes to pack a meal box from <current time> to <target time> by <date>.

Principle 5: Accountability

- Think and act in a manner needed to achieve results
- Hold others responsible for following through on commitments
- Communicate progress
- Capture learning

Project	P	D	S	A	Results
1. Hiring	G	R	G	G	
2. Contracts	Y	G			
3. Permitting	G				
4. Safety	R	G	R		

- **Green** = on schedule,
- **Yellow**= slightly behind schedule
- **Red** = significantly behind schedule

Principle 6: Excellence

Incremental	Systematic	Revolutionary
(local improvement)	(evolutionary)	(breakthrough)
Improving what exists	Distinctly different/better	Radically new & different/better

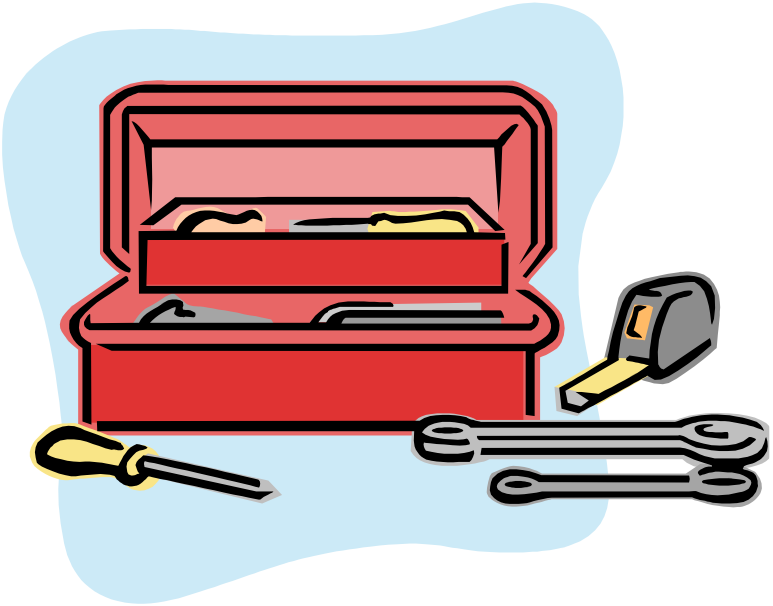




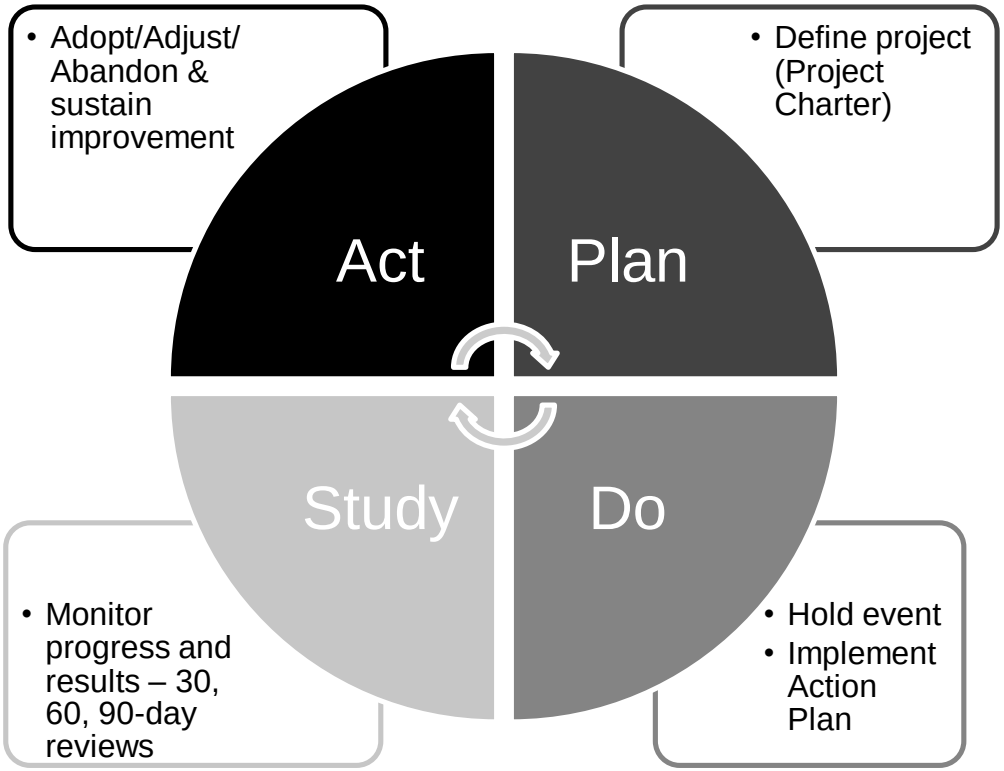


Lean Concepts and Tools

- PDSA
- 7 Wastes
- 5S
- Standard Work
- Visual Management
- Kaizen (Kaizen Event)
- Problem solving



PDSA – The Lean Methodology



Following the Lean methodology ensures knowledge creation and continuous improvement

7 Wastes



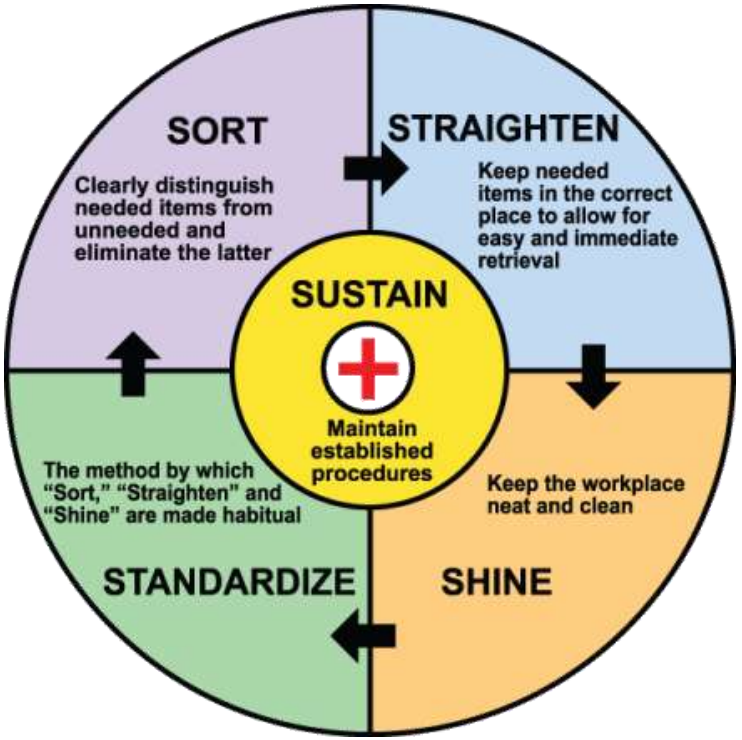
1. Defects
2. Overproduction
3. Waiting
4. Non-utilized staff talent
5. Transportation
6. Inventory
7. Motion
8. Extra processing

5S

A simple method for creating a clean, safe, orderly, high performance work environment.

1. Sort
2. Set In Order
3. Shine
4. Standardize
5. Sustain

6th "S" for "Safety"



Before



5S Numbers Game – Round 1



90 seconds

1S - Sort

"When in doubt, move it out."





“Numbers” – Round 2



60 seconds

2S – Set in Order (Straighten)

"A place for everything, and everything in its place."

A visual management strategy!





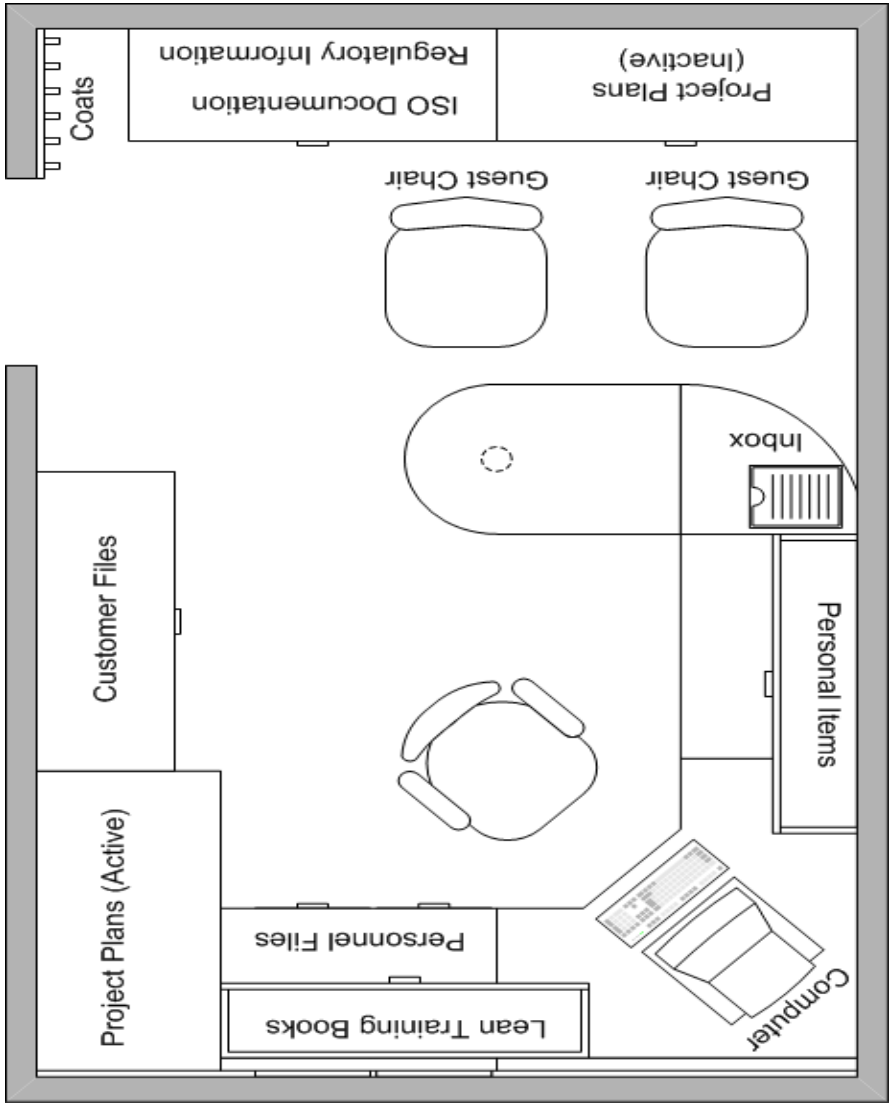
3S - Shine

"The best cleaning is to not need cleaning."



4S - Standardize

"See and recognize what needs to be done."



5S - Sustain

"Effective, ongoing application of 5S"



The most difficult step!

After



5S Tips

- Keep it fun – consider friendly competition
- Leverage teamwork
- Take before and after photos
- Rotate maintaining shared areas among staff
- Provide positive reinforcements



Poka Yoke – Error Proofing



Visual Management

A communication method that tells, at a glance, how work should be done.

- Where items belong
- How many items
- Standard procedure
- Work-in-process (WIP)



There is only one place to put each item.

Communication Boards



Visual management tools that can be understood in 30 seconds or less

Examples: In/Out, project status, staffing, wait times, etc.

Can also communicate accomplishments

Kaizen

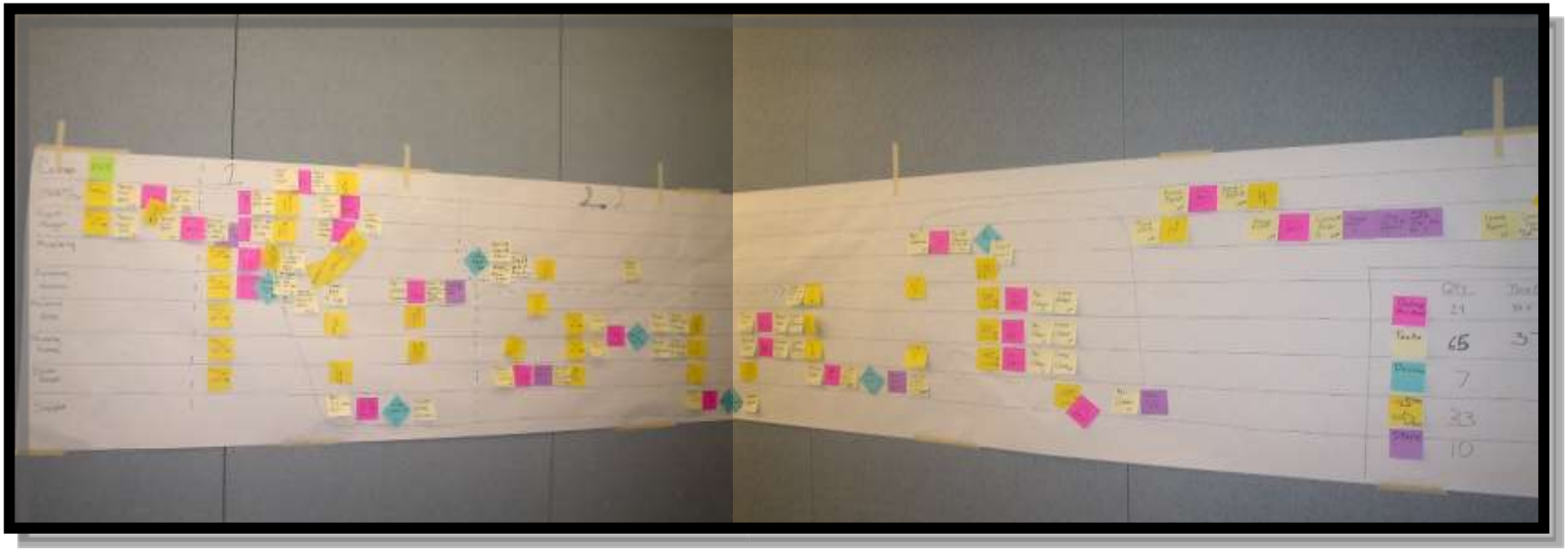
A Kaizen Event is a facilitated, small-scope improvement activity that engages the creativity of employees to reduce waste in a work process. A Kaizen Event typically lasts 3-5 days.



Kaizen, A combination of two Japanese symbols for "change" and "good," most commonly translated as "change for the better."

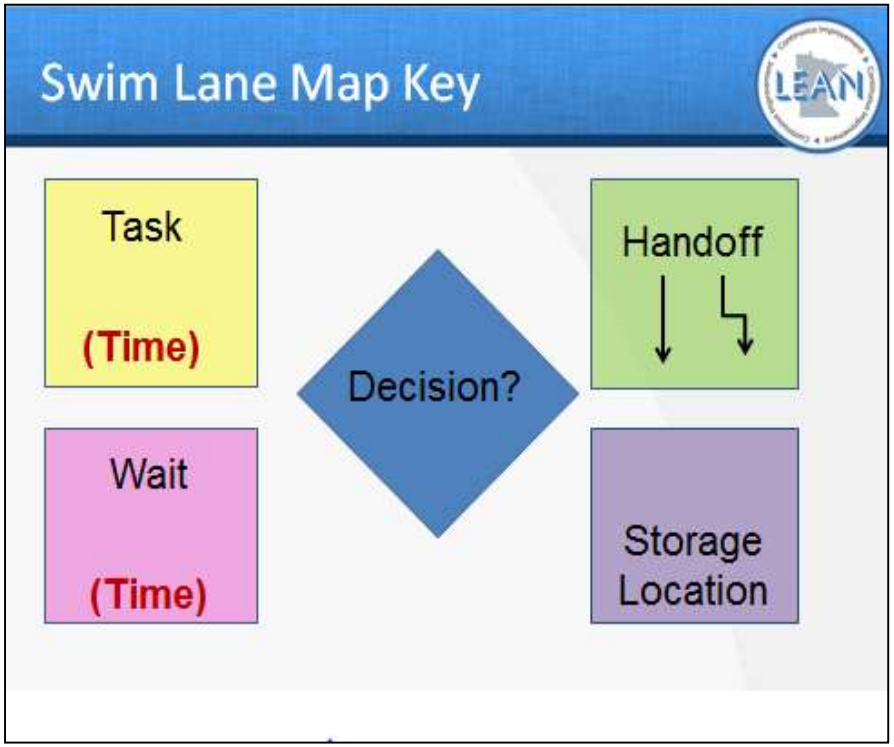
Kaizen Event

Kaizen events use a swim lane map to document the current and future process.



Group Mapping Exercise

- What: Pizza order and delivery process
- Who:
  - Customer
  - Order taker
  - Cook
  - Register attendant
  - Other?



Standard Work

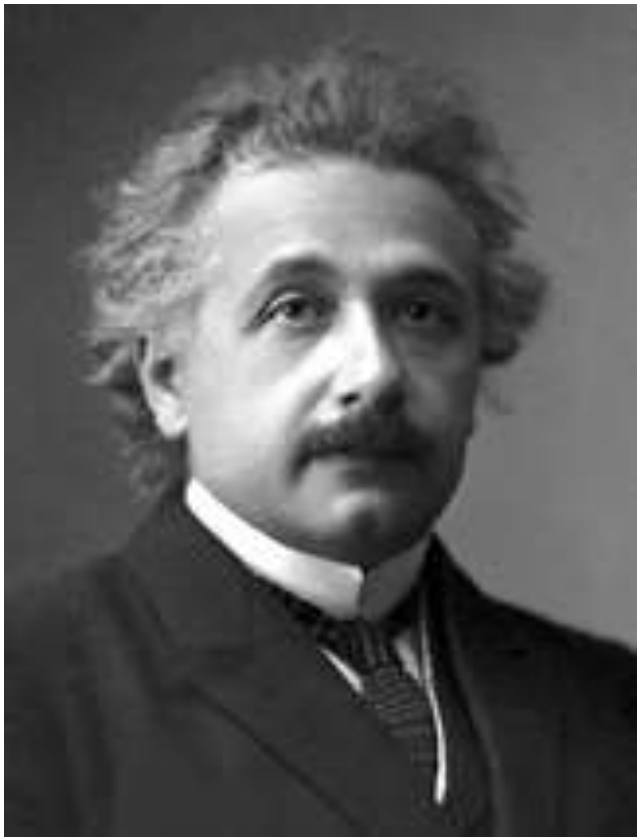
The safest, highest quality, and most efficient way to perform a task or process.

- Focuses on helping the employee be successful
- Reduces variation and increases consistency

"Where there is no standard, there can be no Kaizen."

Taiichi Ohno, Vice-President Toyota Motor Company

Problem Solving



"If I were given one hour to save the world, I would spend 59 minutes defining the problem and one minute solving it."

- Albert Einstein

Problem Solving

Symptoms: You see it, people talk about it; it is visible!

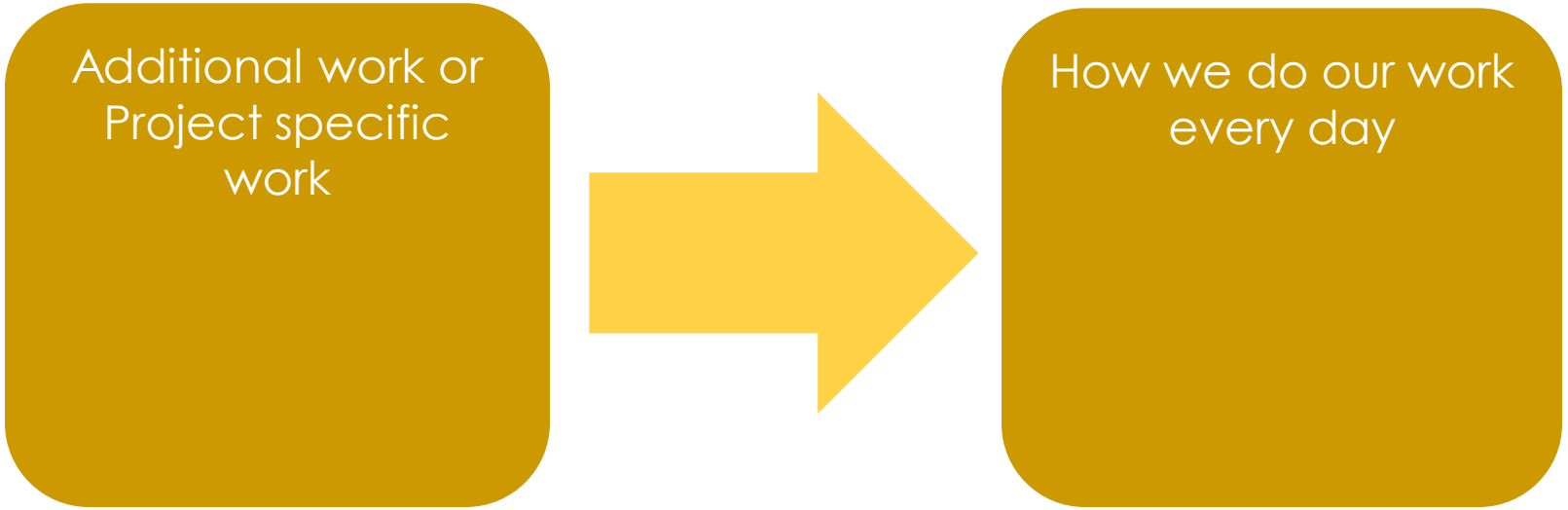


Root Cause: The one to address. It is often hidden. You need to find it!

Defining the “wrong” problem wastes considerable time looking in the wrong direction for solution.

To Create a Lean Culture…

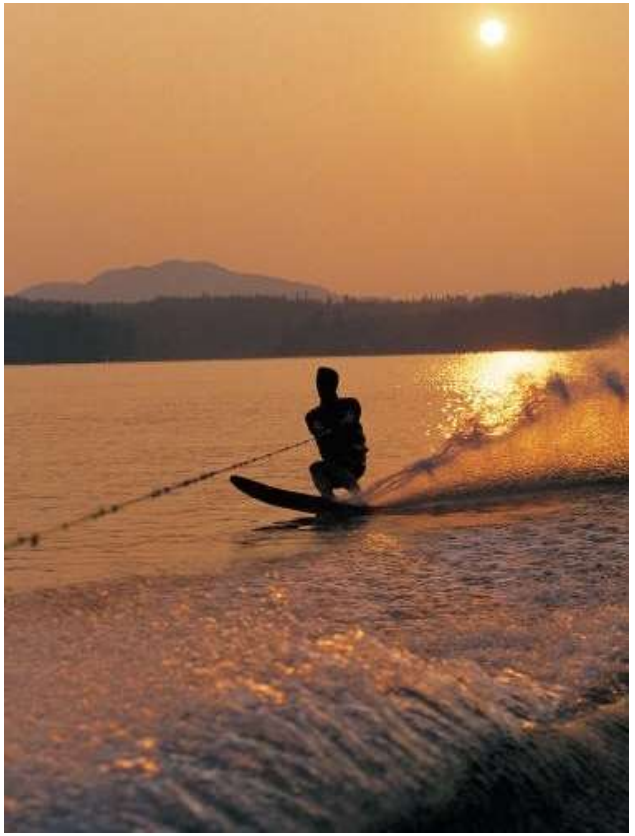
We need to move from viewing Lean as:



Are You Challenging Yourself?

If you don't fall once in a while…

you're not trying hard enough to improve.



Managements' Role

- Model the way
- Challenge the status quo
- Set goals and performance targets
- Align work and dedicate resources
- Engage and empower staff
- Remove barriers
- Build a problem solving culture
- Reward/Recognize high performers

Action You can Take

- Try a tool you learned today!
- 5S your desk, network drive, or common work area
- Create standard work
- Learn more about Lean practices and tools
- Ask your customers what they want
- Think about your goals and how to collect data to start measuring where you are now (so that you can show improvement!)